Referee-style Report on "Matrix Theory Reloaded: A BPS Road to Holography"

(Blair, Lahnsteiner, Obers, Yan)

Summary of the paper. The paper revisits the decoupling limits that define matrix theory on Dp-branes ("matrix p-brane theory", MpT), emphasising that they are BPS limits in which the ten-dimensional target space becomes non-Lorentzian (a p-brane Newton–Cartan-type geometry with a codimension-$(p+1)$ foliation). Three questions are posed and answered: (1) how to classify such limits (via duality orbits of M-theory in the DLCQ); (2) how this non-Lorentzian geometry at asymptotic infinity relates to the Lorentzian AdS-type bulk; (3) how the bulk can be generated intrinsically, which leads to a p-brane generalisation of the $T\bar{T}$ deformation and to new flow equations for DBI actions.

1. Principal achievement

The main achievement is the *geometric identification of the near-horizon limit with the asymptotic BPS decoupling limit.* Writing the Dp-brane near-horizon geometry as $ds^2 = \not{\epsilon}(r)\, dx^A dx_A + \not{\epsilon}(r)^{-1} dx^{A'} dx^{A'}$ with $\not{\epsilon} = (r/\ell)^{(7-p)/2}$, the authors observe that this is the MpT limit parameter ω promoted to a radially dependent function, which is allowed by an emergent dilatation symmetry. Large r then reproduces the flat non-Lorentzian MpT geometry seen by the SYM at infinity. This single observation does a lot of work:

- it clarifies, in an elegant and checkable way, how the Maldacena/IMSY near-horizon limits and the BFSS/matrix-theory limit are the same limit seen on the two sides of the duality, and it answers why a non-Lorentzian space couples to $\mathcal{N} = 4$ SYM while the bulk is Lorentzian;
- it reproduces known cases ($\mathrm{AdS}_5 \times S^5$, IMSY geometries, and $\mathrm{AdS}_3 \times S^3 \times M_4$ from a double D1–D5 limit) and gives a uniform recipe for generating new, non-Lorentzian near-horizon geometries by applying a second limit to off-aligned Dp/Dq systems;
- reading the limit backwards gives a precise sense in which the bulk is a $T\bar{T}$-like deformation of the asymptotic geometry, and yields explicit, new, classical p-brane flow equations, e.g. the general-p equation deforming free scalars to the Dirac–Nambu–Goto action for any p, the three-dimensional flow equation for $p = 2$, and the $p = 1$ equation including the gauge field that explains why plain $T\bar{T}$ does not give the D1 DBI action. These are concrete results that stand independently of the holographic interpretation.

The paper also gives a useful, if largely organisational, synthesis of non-relativistic string theory, matrix string theory, MpT and DLCQ.

2. Principal drawback

The principal drawback is that the *holographic claims are kinematic and classificatory rather than dynamical, and remain unverified.* In particular:

- The headline proposal, "holography = DLCQ^n/DLCQ^m correspondence" with $m > n$, is a conjecture supported by bookkeeping (counting how many harmonic-function "1"s are dropped) in a handful of examples (single-stack Dp, M2/M5, D1–D5). It makes no independent prediction, no observable is matched on both sides, and it is not clear what would falsify it. It is also not clear that it adds content beyond "near-horizon limit = BPS limit at infinity".
- The new non-Lorentzian bulk geometries are obtained by inserting by hand an extra ω-dependent rescaling of the harmonic-function scale (effectively a rescaling of N). The authors acknowledge in the Outlook that they have *not* verified that these geometries solve the equations of motion and torsional constraints of MpT supergravity, that their isometries, supersymmetries and charges are not computed, and that the physical interpretation of the rescaling of N is unclear. Hence the claimed "non-Lorentzian holography" is not yet established as a duality; it is a family of candidate backgrounds.
- The "$T\bar{T}$ deformation generates the bulk" statement relies on formal duality arguments and on promoting a constant coupling to a field-dependent one using a dilatation symmetry. It is not a deformation of the dual field theory in the usual double-trace sense (the authors note this differs from earlier holographic uses of $T\bar{T}$), and the flow equations are classical, bosonic and abelian. For $p = 3$ they are given only to linear order in λ, and for $p > 3$ only at zeroth order, so the $\mathcal{N} = 4$ case that motivates the paper is not fully covered. For $p > 3$ (and generally at asymptotic infinity for $p > 3$) the description also becomes strongly coupled, so large parts of the discussion are restricted to $p \leq 3$.

A secondary issue is presentation: the manuscript is very long and mixes review, new results and conjecture, so it is hard to separate what is derived from what is proposed.

3. Most promising follow-up research idea

Proposal. *Derive the MpT supergravity equations of motion (by taking the $\omega \to \infty$ limit of the type II equations with the critical RR potential, equivalently via worldsheet beta functions) and use them to test and classify the $\not{\leqslant}(r)$ ansatz for non-Lorentzian near-horizon geometries; then compute the isometry algebras of the solutions and match them to the symmetries of the corresponding non-relativistic or foliated field theories.*

Why this one.

- *Feasibility.* It directly addresses the gap the authors themselves flag. The paper already supplies the ansatz (the $T\bar{T}$-deformed MpT geometry written in terms of $\not{\leqslant}$ and $\tilde{\not{\leqslant}}$ in Sec. 5), and the equations of motion follow from a controlled scaling limit of known supergravity, so the first step is a finite, algebraic calculation with a clear success criterion (the ansatz either solves the equations or it does not). Related M2 results in non-relativistic M-theory exist for cross-checks. No new conceptual input is required.
- *Impact.* If the candidates solve the equations, the non-Lorentzian holography programme gets a firm footing; if they do not, the failure shows exactly which constraints (e.g. the torsional ones) fix $\not{\leqslant}(r)$, turning the ansatz into a genuine classification tool for intersecting-brane duals. Computing isometries (and, as the authors suggest, finite temperature) then provides the first non-trivial, falsifiable test of the DLCQ^n/DLCQ^{n+1} picture, because the bulk isometries must match the symmetry of the dual non-relativistic theory.
- *Compared with the alternatives.* Extending the flow equations to the non-abelian or supersymmetric case, for instance whether the $p = 2$ equation applies to BLG/ABJM and flows to

a non-abelian M2 DBI-type action, is attractive, but it is harder (no agreed non-abelian DBI action and subtle ordering ambiguities) and its payoff is more model-specific. It is a good second project once the supergravity side is under control.